

ContactManager

```
1  import java.util.*;
2
3  public class ContactManager {
4      public static void main(String[] args){
5          HashMap<String, Contact> contacts = new HashMap<>();
6
7          contacts.put("Ada Lovelace", new Contact("Ada Lovelace", "+1 617 555
0101"));
8          contacts.put("Grace Hopper", new Contact("Grace Hopper", "+1 617 555
0111"));
9          contacts.put("Frances Allen", new Contact("Frances Allen", "+1 617 555
1000"));
10         contacts.put("Ida Rhodes", new Contact("Ida Rhodes", "+1 617 555
1001"));
11         contacts.put("Katherine Johnson", new Contact("Katherine Johnson", "+1
617 555 1010"));
12
13         if(contacts.get("Ada Lovelace")==null){
14             System.out.println("Contact not found");
15         } else{
16             System.out.println(contacts.get("Ada Lovelace").toString());
17         }
18
19         if(contacts.get("nobody")==null){
20             System.out.println("Contact not found");
21         } else{
22             System.out.println(contacts.get("Ada Lovelace").toString());
23         }
24
25         ArrayList<Contact> sorted = new ArrayList<>(contacts.values());
26         sorted.sort((a, b) -> a.getName().compareTo(b.getName()));
27
28         System.out.println("=== All Contacts ===");
29         for(Contact contact : sorted){
30             System.out.println(contact.toString()+"\n");
31         }
32     }
33 }
34
```